1. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is :

(h = Planck's constant, c = speed of light)

(1) $\frac{2hc}{\lambda}$

(2) $\frac{hc}{3\lambda}$

(3) $\frac{hc}{2\lambda}$

(4) $\frac{hc}{\lambda}$

2. The input signal given to a CE amplifier having a voltage gain of 150 is $V_i = 2\cos\left(15t + \frac{\pi}{3}\right)$. The corresponding output signal will be :

(1) $2\cos\left(15t + \frac{5\pi}{6}\right)$

(2) $300\cos\left(15t + \frac{4\pi}{3}\right)$

(3) $300\cos\left(15t + \frac{\pi}{3}\right)$

(4) $75\cos\left(15t + \frac{2\pi}{3}\right)$

A series R-C circuit is connected to an alternating voltage source. Consider two situations :

(a) When capacitor is air filled.

(b) When capacitor is mica filled.

Current through resistor is i and voltage across capacitor is V then :

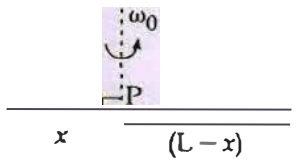
(1) $i_a > i_b$

(2) $V_a = V_b$

(3) $V_a < V_b$

(4) $V_a > V_b$

4. Point masses m_1 and m_2 are placed at the opposite ends of a rigid rod of length L and negligible mass. The rod is to be set rotating about an axis perpendicular to it. The position of point P on this rod through which the axis should pass so that the work required to set the rod rotating with angular velocity ω_0 is minimum, is given by :



(1) $x = \frac{m_2}{m_1}L$

(2) $x = \frac{m_2 L}{m_1 + m_2}$

(3) $x = \frac{m_1 L}{m_1 + m_2}$

(4) $x = \frac{m_1}{m_2}L$

A parallel plate air capacitor has capacity 'C', distance of separation between plates is 'd' and potential difference 'V' is applied between the plates. Force of attraction between the plates of the parallel plate air capacitor is :

(1) $\frac{CV^2}{d}$

(2) $\frac{C^2V^2}{2d^2}$

(3) $\frac{C^2V^2}{2d}$

(4) $\frac{CV^2}{2d}$

6. An ideal gas is compressed to half its initial volume by means of several processes. Which of the process results in the maximum work done on the gas ?

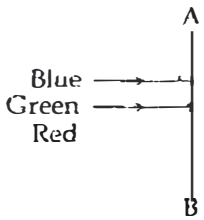
(1) Isochoric

(2) Isothermal

(3) Adiabatic

(4) Isobaric

7. A beam of light consisting of red, green and blue colours is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelength are 1.39, 1.44 and 1.47, respectively.



The prism will :

(1) not separate the three colours at all

(2) separate the red colour part from the green and blue colours

(3) separate the blue colour part from the red and green colours

(4) separate all the three colours from one another

8. Two vessels separately contain two ideal gases A and B at the same temperature, the pressure of A being twice that of B. Under such conditions, the density of A is found to be 1.5 times the density of B. The ratio of molecular weight of A and B is :

(1) 2

(2) 2

(3) $\frac{1}{3}$

(4) 4

A remote - sensing satellite of earth revolves in a circular orbit at a height of 0.25×10^6 m above the surface of earth. If earth's radius is 6.38×10^6 m and $g = 9.8\ ms^{-2}$, then the orbital speed of the satellite is :

(1) 9.13 km s^{-1}

(2) 6.67 km s^{-1}

(3) 7.76 km s^{-1}

(4) 8.56 km s^{-1}

10. The energy of the em waves is of the order of 15 keV. To which part of the spectrum does it belong ?

(1) Ultraviolet rays

(2) γ - rays

(3) X - rays

(4) Infra - red rays

11. A proton and an alpha particle both enter a region of uniform magnetic field B, moving at right angles to the field B. If the radius of circular orbits for both the particles is equal and the kinetic energy acquired by proton is 1 MeV, the energy acquired by the alpha particle will be :

(1) 1.5 MeV

(2) 1 MeV

(3) 4 MeV

(4) 0.5 MeV

12. If vectors $\vec{A} = \cos\omega t\ \hat{i} + \sin\omega t\ \hat{j}$ and $\vec{B} = \cos\left(\frac{\omega t}{2}\right)\hat{i} + \sin\frac{\omega t}{2}\ \hat{j}$ are functions of time, then the value of t at which they are orthogonal to each other is :

(1) $t = \frac{\pi}{\omega}$

(2) $t = 0$

(3) $t = \frac{\pi}{4\omega}$

(4) $t = \frac{\pi}{2\omega}$

13. A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m^2. The coil carries a current of 2 A. If the plane of the coil is inclined at an angle of 30° with the direction of the field, the torque required to keep the coil in stable equilibrium will be :

(1) 0.24 Nm

(2) 0.12 Nm

(3) 0.15 Nm

(4) 0.20 Nm

14. An automobile moves on a road with a speed of 54 km h^{-1}. The radius of its wheels is 0.45 m and the moment of inertia of the wheel about its axis of rotation is 3 kg m^2. If the vehicle is brought to rest in 15 s, the magnitude of average torque transmitted by its brakes to the wheel is :

(1) 10.86 kg $m^2 s^{-2}$

2.86 kg $m^2 s^{-2}$

6.66 kg $m^2 s^{-2}$

8.58 kg $m^2 s^{-2}$

15. Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the metal wires respectively, the effective conductivity of the combination is :

(1) $\frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$

(2) $\sigma_1 + \sigma_2$

(3) $\frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$

(4) $\frac{\sigma_1 + \sigma_2}{2\sigma_1 \sigma_2}$

16. If potential (in volts) in a region is expressed as $V(x, y, z) = 6xy - y + 2yz$, the electric field (in N/C) at point $(1, 1, 0)$ is :

(1) $-\left(2\hat{i} + 3\hat{j} + \hat{k}\right)$

(2) $-\left(6\hat{i} + 9\hat{j} + \hat{k}\right)$

(3) $-\left(3\hat{i} + 5\hat{j} + 3\hat{k}\right)$

(4) $-\left(6\hat{i} + 5\hat{j} + 2\hat{k}\right)$

17. Two particles A and B, move with constant velocities $\vec{v_1}$ and $\vec{v_2}$. At the initial moment their position vectors are $\vec{r_1}$ and $\vec{r_2}$ respectively. The condition for particles A and B for their collision is :

(1) $\vec{r}_1 \times \vec{v}_1 = \vec{r}_2 \times \vec{v}_2$

(2) $\vec{r}_1 - \vec{r}_2 = \vec{v}_1 - \vec{v}_2$

(3) $\frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = \frac{\vec{v}_2 - \vec{v}_1}{|\vec{v}_2 - \vec{v}_1|}$

(4) $\vec{r}_1 \cdot \vec{v}_1 = \vec{r}_2 \cdot \vec{v}_2$

18. 4.0 g of a gas occupies 22.4 litres at NTP. The specific heat capacity of the gas at constant volume is 5.0 $JK^{-1} mol^{-1}$. If the speed of sound in this gas at NTP is 952 ms^{-1}, then the heat capacity at constant pressure is

(Take gas constant $R = 8.3\ JK^{-1} mol^{-1}$)

(1) 7.0 $JK^{-1} mol^{-1}$

(2) 8.5 $JK^{-1} mol^{-1}$

(3) 8.0 $JK^{-1} mol^{-1}$

(4) 7.5 $JK^{-1} mol^{-1}$

19. A force $F = \alpha i + 3j + 6k$ is acting at a point $r = 2i - 6j - 12k$. The value of α for which angular momentum about origin is conserved is :

(1) zero

(2) 1

(3) -1

(4) 2

20. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern, the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is :

(1) π radian

(2) $\frac{\pi}{8}$ radian

(3) $\frac{\pi}{}$ radian

(4) $\frac{}{2}$ radian

21. The heart of a man pumps 5 litres of blood through the arteries per minute at a pressure of 150 mm of mercury. If the density of mercury be $13.6 \times 10^3\ kg/m^3$ and $g = 10\ m/s^2$ then the power of heart in watt is :

(1) 3.0

(2) 1.50

(3) 1.70

(4) 2.35

22. A ball is thrown vertically downwards from a height of 20 m with an initial velocity v_0. It collides with the ground, loses 50 percent of its energy in collision and rebounds to the same height. The initial velocity v_0 is : (Take $g = 10\ ms^{-2}$)

(1) 28 ms^{-1}

(2) 10 ms^{-1}

(3) 14 ms^{-1}

(4) 20 ms^{-1}

23. The cylindrical tube of a spray pump has radi R, one end of which has n fine holes, each of i r If the speed of the liquid in the tube is V, the speed of the ejection of the liquid through the holes i

(1) $\frac{VR^2}{n^3r^2}$

(2) $\frac{V^2R}{nr}$

(3) $\frac{VR^2}{n r}$

(4) $\frac{VR^2}{nr^2}$

24. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequency for this string is :

(1) 10.5 Hz

(2) 105 Hz

(3) 155 Hz

(4) 205 Hz

If dimensions of critical velocity v_c of a liquid flowing through a tube are expressed as $[\eta^x \rho^y r^z]$, where η, ρ and r are the coefficient of viscosity of liquid, density of liquid and radius of the tube respectively, then the values of x, y and z are given by :

(1) $-1, -1, -1$

(2) $1, 1, 1$

(3) $1, -1, -1$

(4) $-1, -1, 1$

26. A nucleus of uranium decays at rest into nuclei of

(1) thorium and helium. Th n : than the thorium nucleus.

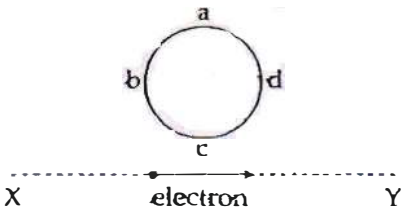
(2) The helium nucleus has less kinetic energy than the thorium nucleus.

(3) The helium nucleus has more kinetic energy than the thorium nucleus.

(4) The helium nucleus has less momentum the thorium nucleus

(1) The helium nucleus has more momentum

27. An electron moves on a straight line path XY as shown. The abcd is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil ?



(1) The current will reverse its direction as the electron goes past the coil

(2) No current induced

(3) abcd

(4) adcb

28. Water rises to a height 'h' in capillary tube. If the length of capillary tube above the surface of water is made less than 'h', then :

(1) water rises upto a point a little below the top and stays there.

(2) water does not rise at all.

(3) water rises upto the tip of capillary tube and then starts overflowing like a fountain.

water rises upto the top of capillary tube and stays there without overflowing.

29. In an astronomical telescope in normal adjustment a straight black line of length L is drawn on inside part of objective lens. The eye-piece forms a real image of this line. The length of this image is I. The magnification of the telescope is :

(1) $\frac{L+I}{L-I}$

(2) $\frac{L}{I}$

(3) $\frac{L}{I}+1$

(4) $\frac{L}{I}-1$

30. A circuit contains an ammeter, a battery of 30 V and a resistance 40.8 ohm all connected in series. If the ammeter has a coil of resistance 480 ohm and a shunt of 20 ohm, the reading in the ammeter will be :

(1) 2 A

(2) 1 A

(3) 0.5 A

(4) 0.25 A

31. On a frictionless surface, a block of mass M moving at speed v collides elastically with another block of same mass M which is initially at rest. After collision the first block moves at an angle θ to its initial direction and has a speed $\frac{v}{3}$. The second block's speed after the collision is :

(1) $\frac{\sqrt{2}}{3}v$

(2) $\frac{\sqrt{3}}{4}v$

(3) $\frac{2\sqrt{2}}{3}v$

(4) $\frac{3}{4}v$

32. A satellite S is moving in an elliptical orbit around the earth. The mass of the satellite is very small compared to the mass of the earth. Then,

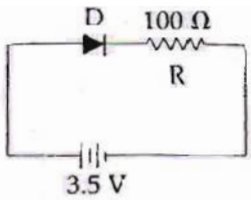
(1) the linear momentum of S remains constant in magnitude.

(2) the acceleration of S is always directed towards the centre of the earth.

(3) the angular momentum of S about the centre of the earth changes in direction, but its magnitude remains constant.

(4) the total mechanical energy of S varies periodically with time.

33. In the given figure, a diode D is connected to an external resistance R = 100 Ω and an e.m.f. of 3.5 V. If the barrier potential developed across the diode is 0.5 V, the current in the circuit will be :



(1) 20 mA

(2) 35 mA

(3) 30 mA

(4) 40 mA

34. A potentiometer wire of length L and a resistance r are connected in series with a battery of e.m.f. E_0 and a resistance r_1. An unknown e.m.f. E is balanced at a length l of the potentiometer wire. The e.m.f. E will be given by :

(1) $\frac{E_0\, l}{L}$

(2) $\frac{L\, E_0\, r}{(r + r_1)l}$

(3) $\frac{L\, E_0\, r}{l\, r_1}$

(4) $\frac{E_0\, r}{(r + r_1)\quad L}$

35. Two stones of masses m and 2 m are whirled in horizontal circles, the heavier one in a radius $\frac{r}{2}$ and the lighter one in radius r. The tangential speed of lighter stone is n times that of the value of heavier stone when they experience same centripetal forces. The value of n is :

(1) 4

(2) 1

(3) 2

(4) 3

36. Two slits in Youngs experiment have widths in the ratio 1 : 25. The ratio of intensity at the maxima and minima in the interference pattern, $\frac{I_{max}}{I_{min}}$ is :

(1) $\frac{49}{121}$

(2) $\frac{4}{9}$

(3) $\frac{9}{4}$

(4) $\frac{121}{49}$

37. The Young's modulus of steel is twice that of brass. Two wires of same length and of same area of cross section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of :

(1) 4 : 1

(2) 1 : 1

(3) 1 : 2

(4) 2 : 1

38. The coefficient of performance of a refrigerator is 5. If the temperature inside freezer is $-20°C$, the temperature of the surroundings to which it rejects heat is :

(1) 11°C

(2) 21°C

(3) 31°C

(4) 41°C

39. Light of wavelength 500 nm is incident on a metal with work function 2.28 eV. The de Broglie wavelength of the emitted electron is :

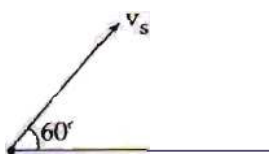
(1) $\geq 2.8 \times 10^{-9}$ m

(2) $\leq 2.8 \times 10^{-12}$ m

(3) $< 2.8 \times 10^{-10}$ m

(4) $< 2.8 \times 10^{-9}$ m

40. A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 19.4 ms^{-1} at an angle of 60° with the source observer line as shown in the figure. The observer is at rest. The apparent frequency observed by the observer (velocity of sound in air 330 ms^{-1}), is :



(1) 106 Hz

(2) 97 Hz

(3) 100 Hz

(4) 103 Hz

41. The value of coefficient of volume expansion of glycerin is $5 \times 10^{-4} K^{-1}$. The fractional change in the density of glycerin for a rise of 40°C in its temperature, is :

(1) 0.025

(2) 0.010

(3) 0.015

(4) 0.020

e position vector of a particle R as a function of ti e is given by :

$\vec{R} = 4 \sin(2\pi t)\, \hat{i} + 4 \cos(2\pi t)\, \hat{j}$

Where R is in meters, t is in seconds and $\hat{i}$ and $\hat{j}$ denote unit vectors along x-and y-directions, respectively. Which one of the following statements is **wrong** for the motion of particle ?

(1) Magnitude of the velocity of particle is 8 meter/second

(2) Path of the particle is a circle of radius 4 meter.

(3) Acceleration vector is along $-R$

(4) Magnitude of acceleration vector is $\frac{v^2}{R}$, where v is the velocity of particle.

43. A plank with a box on it at one end is gradually raised about the other end. As the angle of inclination with the horizontal reaches 30°, the box starts to slip and slides 4.0 m down the plank in 4.0 s. The coefficients of static and kinetic friction between the box and the plank will be, respectively :



(1) 0.5 and 0.6

(2) 0.4 and 0.3

(3) 0.6 and 0.6

(4) 0.6 and 0.5

44. In the spectrum of hydrogen, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series is :

(1) $\frac{27}{5}$

(2) $\frac{5}{27}$

(3) $\frac{4}{9}$

$\frac{9}{4}$

45. A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then, its time period of vibration will be :

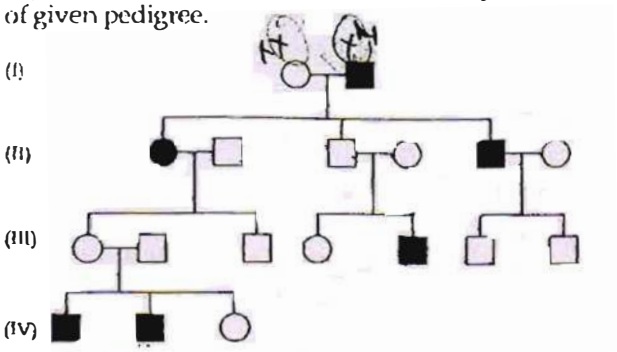
(1) $\frac{\beta^2}{\alpha}$

(2) $\frac{2\pi\beta}{\alpha}$

(3) $\frac{\beta^2}{\alpha^2}$

(4) $\frac{\alpha}{\beta}$

46. In the following human pedigree, the filled symbols represent the affected individuals. Identify the type of given pedigree.



(1) Autosomal recessive

(2) X-linked dominant

(3) Autosomal dominant

(4) X-linked recessive

47. Which of the following is **not** a function of the skeletal system ?

(1) Production of body heat

(2) Locomotion

(3) Production of erythrocytes

(4) Storage of minerals

Destruction of the anterior horn cells of the spinal cord would result in loss of :

(1) commissural impulses

(2) integrating impulses

(3) sensory impulses

(4) voluntary motor impulses

49. The term "linkage" was coined by

(1) G. Mendel

(2) W. Sutton

(3) T.H. Morgan

(4) T. Boveri

50. Filiform apparatus is characteristic feature of

(1) Aleurone cell

(2) Synergids

(3) Generative cell

(4) Nucellar embryo